

Ziyi Zhuang

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ACADEMIC BACKGROUND

The University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Mathematics

September 2022 - April 2026

- Relevant Courses: Probability, Stochastic Processes, Linear Programming, Game Theory, Differential Equations, Mathematical Modelling, Applied Machine Learning
- GPA: 80.9 / 100

TECHNICAL SKILLS

Mathematics & Statistics: Probability, Stochastic Processes, Statistical Inference, Optimization, Game Theory, Dynamical Systems

Data & Computing: Python, MATLAB, Excel, Regression Analysis, Time-Series Analysis, A/B Testing, Machine Learning

Languages: English (Fluent), Mandarin (Fluent)

RESEARCH EXPERIENCE

Game-Theoretic Approach to Nash Equilibria in Network Systems

Math 444 - Supervisor: Prof. Nahid Walji

- Modelled transportation and wireless networks as non-cooperative games; derived Nash equilibria and quantified inefficiencies using the Price of Anarchy.
- Established an upper bound of 2 for traffic networks and ~ 0.85 for wireless allocation models.
- Proposed coordination strategies (e.g., link removal, optimal flow) to align individual and system-wide optima.

Comparative Modelling of Delta-Notch Signalling Pathways

Math 462 - Supervisor: Prof. Eric Cytrynbaum

- Compared ODE, SDE, and agent-based models for cell differentiation dynamics in biological signalling systems; proved convergence of stochastic to deterministic models.
- Conducted stability and sensitivity analyses to assess the impact of noise and parameter variation on pattern formation.
- Strengthened expertise in dynamical systems, stochastic processes, and computational biology.

PROFESSIONAL & APPLIED EXPERIENCE

NFLS British Columbia Academy

Nanjing, China

Teaching Assistant & Data Support

May 2025 - June 2025

- Delivered 100+ hours of classroom and tutoring support for Grade 10–12 mathematics, improving student performance by 3% on average.
- Maintained and analyzed student performance data using Office Automation platforms; developed data-driven reports to support enrollment and placement strategies.
- Recognized for precision, initiative, and analytical support in both instructional and operational roles.

Jiangsu Bozhan - Museum Exhibition Design Co., Ltd.

Nanjing, China

Media Platform Data Analyst

May 2024 - August 2024

- Developed quantitative models in Python and Excel to forecast user engagement and optimize content distribution, increasing online exposure by 23% in three months.
- Applied regression and time-series analysis to identify key drivers of traffic; optimized posting schedules, improving reach by 8%.
- Designed and implemented A/B testing to evaluate creative and scheduling strategies, improving conversion rate by 15%.
- Utilized stochastic processes and machine learning (MATH 303, CPSC 330) for predictive modelling and risk-aware media planning.

School of Environment, Nanjing Normal University

Nanjing, China

Research Intern - Environmental Data Modelling

January 2023

- Processed and analyzed 7-year air quality datasets using WRF-Chem and epidemiological models.
- Implemented damage function models to estimate PM_{2.5} and O₃-related mortality; conducted sensitivity analysis on urbanization impact.
- Automated data processing and visualization using Python.